

10 Real-Time Questions and Answers from the Workforce Dataset

Based on the **ABC Constructions Workforce Attendance, Skills, Overtime, Safety and Insurance Dataset** containing **1,500 records** for **50 workers** over **30 days**:

1. How many workers are available on the construction site?

Answer: There are **50 workers** in total.

2. How many workers are assigned to overtime work?

Answer: **20 workers** are assigned to overtime work.

$$\{20\}/\{50\} \times 100 = 40\%$$

Therefore, **40% of the workforce** is assigned to overtime.

3. How many Sundays are recorded as weekly holidays?

Answer: The dataset covers 30 days and contains **4 Sundays**. Since there are 50 workers:

$$4 \times 50 = 200$$

Therefore, **200 attendance records** are marked as weekly holidays (H).

4. What is the overall attendance rate on working days?

Answer: There are **1,300 worker-day records** excluding Sundays. The overall attendance rate is approximately:

$$96.6\%$$

This indicates good workforce availability.

5. How many worker absences were recorded?

Answer: The dataset contains **44 absence records** during the working days.

This represents approximately:

$$\{44\}/\{1300\} \times 100 = 3.38\%$$

of all scheduled working-day worker records.

6. Which skill category has the highest number of workers?

Answer: Concrete Work has the highest number of workers, with **8 workers**.

The workforce distribution includes:

Skill Category	Workers
Concrete Work	8
Masonry	7
Carpentry	7
Steel Fixing	7
General Labour	6
Equipment Operation	5
Plumbing	5
Electrical Work	4
Survey Assistance	1

7. Which skill category has the highest number of overtime workers?

Answer: General Labour has the highest number of overtime workers, with **6 workers**.

This indicates that material handling, site cleaning, scaffolding assistance, and construction support activities require additional labour during overtime periods.

8. How many total overtime hours were recorded?**Answer:** The dataset records a total of:

1,509 overtime hours

The difference from the theoretical maximum occurs because some overtime workers were absent on certain working days.

9. What is the total overtime payout?

Answer: At an overtime rate of **₹150 per hour**, the total overtime payout is:

$1,509 \times ₹150 = ₹2,26,350$

Therefore, the company should budget approximately **₹2.26 lakh** for the recorded overtime work.

10. Which worker had the highest number of absences?

Answer: Worker W-029 recorded the highest number of absences, with **2 absence records** during the 30-day period.

The project manager should review the attendance history of workers with repeated absences to determine whether the reason is related to:

- Health
- Transport
- Personal leave
- Fatigue